# A Signal is Not an Edge:

## *Post-Mortem of a Bitcoin Prediction-Market Strategy*

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## I. Abstract

Prediction markets like Kalshi directly expose something that is implied or hidden in most markets: probabilities. In that regard, looking at Kalshi's BTC markets was an interesting case. My original hypothesis was that a retail-dominated market could have produced miscalibrated probabilities, especially when compared to more industrial markets, such as the options market. So, instead of looking at individual trades, I decided to treat Kalshi's entire BTC contract market as a probabilistic model, scoring and evaluating it as one.

The project eventually grew into a full research pipeline. I collected minute-resolution Kalshi prices, hourly settlement records, and Binance market data, then restricted the contracts to strikes near the money. Using this dataset, I evaluated Kalshi's forecast performance across market regimes and tested increasingly specialized independent models against both Kalshi's probabilities and realized trading outcomes.

## II. Kalshi's Market as a Probabilistic Forecasting Model

The first step in my research was to test my initial hypothesis and evaluate just how efficient Kalshi's market is. My first dataset used 218,098 scored contract-minute forecasts across 3,669 event contracts, corresponding to 1,223 hourly events – with three contracts per event, using the market's stored YES price as its probability forecast, and the official settlement as the binary outcome. Because each contract appears once per minute, these are repeated forecasts rather than 218,098 completely independent events, which is important to note when considering the sample size.

The market proved a strong benchmark. Its Brier score was 0.126806, its log loss was 0.388344, and its expected calibration error (ECE) was 0.010594. The last metric signifies that, across calibration bins, the weighted average absolute difference between predicted and observed frequencies was about 1.06 percentage points – but not necessarily that every probability bin was accurate within 1 point.

| Brier | Log loss | ECE |
|---|---|---|
| 0.126806 | 0.388344 | 0.010594 |

The contract-minute bin carried even more information. Brier score fell from 0.1952 during the first 1-10 minutes of an hourly contract to 0.0361 during the final 51-60 minutes. Resolution also rose from about 0.054 to 0.214, while reliability error remained very small. In plain language, the market became significantly more decisive as settlement approached, without losing its overall calibration. This finding largely rejected my original hypothesis; there was no obvious global failure in Kalshi's market pricing that a generic model could easily exploit.

## III. Kalshi's Performance Degradation During High Volatility Hours

The next question I asked was conditional. If Kalshi's market performs well on a global scale, when did the market's probability forecasts perform worse? After examining several conditional slices, I computed each hour's RV from 60 one-minute Binance bars and labeled corresponding hours a "high volatility extreme" when its RV was at or above the rolling 90$^{th}$ percentile of a 72-hour window. On the later data snapshot, the total comparison contained 226,149 contract-minute rows across 1,268 hours. The extreme slice itself contained 154 hours and 462 contracts.

Within that slice, Kalshi's Brier score rose from 0.1269 to 0.1533, a 20.8% relative deterioration. Log loss also increased, from 0.3866 to 0.4546, and ECE increased from 0.0107 to 0.0449. This is evidence of a real deterioration in conditional forecast scores on HVE hours. The event mix and base rates changed, and, most importantly, the extreme-volatility label used the completed hour's RV. The label was an ex-post diagnostic, not information that was available during the hour – hence not a live trading signal.

| Metric | Brier | Log Loss | ECE |
|---|---|---|---|
| Global | 0.1269 | 0.3866 | 0.0107 |
| Extreme Volatility | 0.1533 | 0.4546 | 0.0449 |
| Delta | 0.0264 | 0.0680 | +0.0342 |
| Relative Change (%) | +20.8% | +17.6% | +319.6% |

## IV. Building an Independent Probability Model

To create an independent benchmark to compare with Kalshi's market, I built a Student-t GARCH(1,1) Monte Carlo model. Its conditional variance followed the familiar update $h(t+1) = \omega + \alpha\varepsilon(t)^2 + \beta h(t)$. The Student-t allowed for heavier tails than a Gaussian model. At each forecast time, the model simulated the remaining path to the end of the hour (60 – t steps) and estimated a YES probability as the fraction of terminal prices above the contract strike.

The committed baseline used two days of one-minute returns, refit its parameters every 60 forecast minutes, updated variance recursively between fits, and generated 500 antithetic terminal paths per forecast. On the same global 218,098 rows, its Brier score was 0.1338, log loss was 0.4093, and ECE was 0.0159. The model was sufficiently close to provide a useful independent benchmark and sanity check for the evaluation pipeline.

| Model | Brier | Log loss | ECE |
|---|---|---|---|
| Kalshi | 0.1268 | 0.3883 | 0.0106 |
| Independent P | 0.1338 | 0.4093 | 0.0159 |
| Delta | +0.007 | +0.0210 | +0.0053 |

Next, I built a volatility-specialized version. This version was fitted by maximum likelihood on historical minutes in the extreme-volatility regime, used 10,000 Monte Carlo paths, and added a post-hoc isotonic calibration layer trained only on pre-backtest historical examples. On the exact HVE overlap, the model reduced ECE from Kalshi's 0.0449 to 0.0318 – about a 29% improvement. Simultaneously, its Brier score was 0.1560 versus 0.1539 and its log loss was 0.4767 versus 0.4546. The model produced a material improvement in ECE while modestly worsening on other metrics. This illustrates how better calibration on one summary metric did not make the forecast better overall.

| Model | Brier | Log loss | ECE |
|---|---|---|---|
| Kalshi | 0.1539 | 0.4546 | 0.0449 |
| Volatility Specialized | 0.1560 | 0.4767 | 0.0318 |
| Delta | +0.0021 | +0.0221 | -0.0131 |
| Relative Change % | 1.4% | 4.9% | -29.2% |

## V. Reframing the Question

While the volatility-specialized model produced promising results, there was still a caveat that needed to be solved before applying it to backtests against the market. The prior volatility-specialized version only exhibits that performance in the high volatility extreme minutes – which need to be forecasted ahead of time. The solution I came up with was to use a causal gate – a volatility outbreak probabilistic model, using high-frequency, minute-level Binance futures features, including: rolling returns, RV, aggregate trade flow and imbalance, funding and premium variables, and percentage-bucketed depth snapshots. The target was precise: a row anchored at minute t predicted whether minute t+1 would be a range breakout, defined as $\ln\left(\frac{high}{low}\right)$ exceeding prior 72-minute $90^{th}$ percentile after three calm candles. Training used chronological 30-day windows, followed by 7 days for validation and calibration, as well as 7 days for testing, rolling forward by 7 days. There was no random split, and the test window was not used for fitting or calibration. In the recorded run, the highest-risk decile had 15.99% breakout precision against an 8.06% eligible base rate. That is 1.98× lift, with a PR-AUC of 0.1286.

The final probability model used the classifier output *q* as a continuous weight. The output *q* measures the probability of a next minute range breakout, which the strategy uses to answer: "How much should we trust the HVE model?" To prevent bleed during non-breakout minutes, I used a hybrid of the base model and the volatility-specialized model. The exponentially weighted hybrid used:

$$P_{Final} = \big(1 - w(q)\big)P_{Normal} + w(q)P_{Shock}$$

Where:

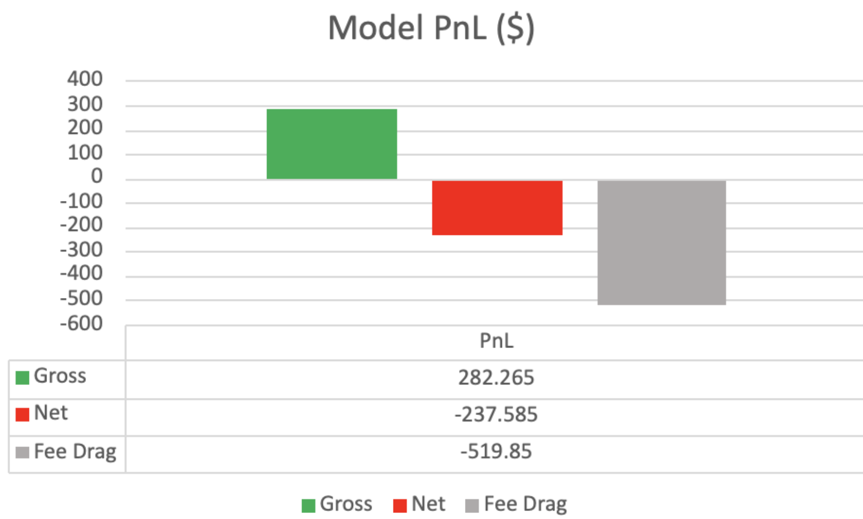
$$w(q) = \frac{e^{2q} - 1}{e^{2} - 1}$$

A separate HAR-RV model forecast hourly volatility from previous hours and its trailing 6, 24, and 72-hour averages. This served as a regime gate where trading was active only when that causal forecast exceeded the historical $75^{th}$ percentile. Still, the one-minute breakout target is not the same event as the hour-level realized volatility regime used to specialize the shock model. The hybrid is, therefore, a heuristic mixture of experts, not a formally aligned posterior probability that the regime is active.

## VI. The PnL Backtest and Signal vs. Edge

A historical simulation backtest that used midpoint prices, idealized fills, and fees modeled according to Kalshi's documentation during the simulation window produced a small gain before costs, but modeled fees turned that gain into a loss. The model selected one YES or NO contract on each qualifying contract-minute row. Qualifying minute rows were defined as having a sufficient edge/difference against Kalshi's market probability after fees. Additionally, a q75 HAR-RV filter was used as an activation gate. Gross PnL was $282.265, with modeled fee drag being $519.850, leaving net PnL at –$237.585. At the same simulated entries, fees would have needed to be reduced roughly 45.7% for the strategy to break even. Gross PnL averaged approximately $0.0083 per entry; modeled fees averaged $0.0154 per entry, producing a net loss per entry of $0.0070.

The reported 33,854 observations are separate one-contract, minute-level counterfactual entries from May 4 through June 30, 2026 UTC, selected during only q75 HAR-RV-active hours. Repeated signals in the same contract were counted as separate counterfactual entries, and valued as though held to official settlement. Entry prices use historical YES mid-price proxies—not executable asks—with NO prices constructed as one minus the YES proxy. The analysis does not model fills, slippage, liquidity, capital constraints, inventory limits, overlapping exposure, or portfolio cash flows; summed PnL should therefore not be interpreted as the performance of an executable portfolio.




Modeled strategy PnL before and after fees.

## VII. Final Remarks and Conclusions

If I were to continue this project, I would begin by replacing the historical midpoint prices with executable market data. This would require collecting timestamped bids, asks, displayed volume, and order-book depth, then defining realistic assumptions for decision latency, queue position, fill probability, slippage, and missed trades. The original simulation lost money after modeled fees alone; incorporating spreads and imperfect execution would likely make the result worse. Consequently, the current strategy should not be treated as deployable until it passes an executable-price test.

I would also investigate whether the gross simulated advantage is concentrated in a smaller subset of trades. A future policy could require the model probability to exceed the market's fee-adjusted break-even probability by a conservative margin, while also passing calibration, reliability, liquidity, and uncertainty thresholds. This would not guarantee profitability, but it could test whether lower-confidence trades were creating fee drag while a smaller group retained meaningful expected value.

This project began with a reasonable hypothesis: Bitcoin volatility might create periods in which Kalshi's contract prices become less reliable, allowing a specialized forecasting system to identify temporary mispricing. The investigation found evidence of predictive signal, but it did not establish an executable trading edge. The volatility classifier produced meaningful ranking power, with precision in its highest-scored decile reaching 15.99% against an 8.06% base rate. The probability models also revealed measurable differences in performance during retrospectively identified high-volatility periods. However, those findings did not translate into superior market forecasts, and some of the strongest conditional results depended on information not available when a live decision would be made.

The final historical simulation exposed the largest gap between prediction and execution. Across 33,854 simulated one-contract trades, the strategy's gross PnL was approximately $282.27, but incurred $519.85 in modeled fees, producing a net loss of $237.59. This result was based on midpoint-price proxies rather than guaranteed executable bids or asks. Fees therefore eliminated the gross advantage before accounting for spread, limited depth, queue position, latency, slippage, adverse selection, or missed fills. This does not prove that no profitable strategy could be developed from the signal. It shows that the evidence collected through this project was insufficient to support that claim.

The model detected structure, but the structure was either too weak, too conditionally defined, or too expensive to trade under the simulated policy. The most valuable result of the project was therefore not a profitable strategy, but a clearer standard for what constitutes one. A useful classifier is not automatically an accurate probability model. An accurate probability model is not automatically superior to the market. A statistical advantage over the market is not automatically large enough to survive execution costs. By following the idea from volatility detection through probability estimation and finally into fee-aware simulation, I built a process capable of rejecting its own hypothesis. Rather than stopping when the early results appeared promising, I continued until the proposed edge encountered the economic constraints of actual trading. The strategy was not validated for deployment, but the research process successfully identified why—and that distinction is the central lesson of the project.

## Appendix A: Source Map

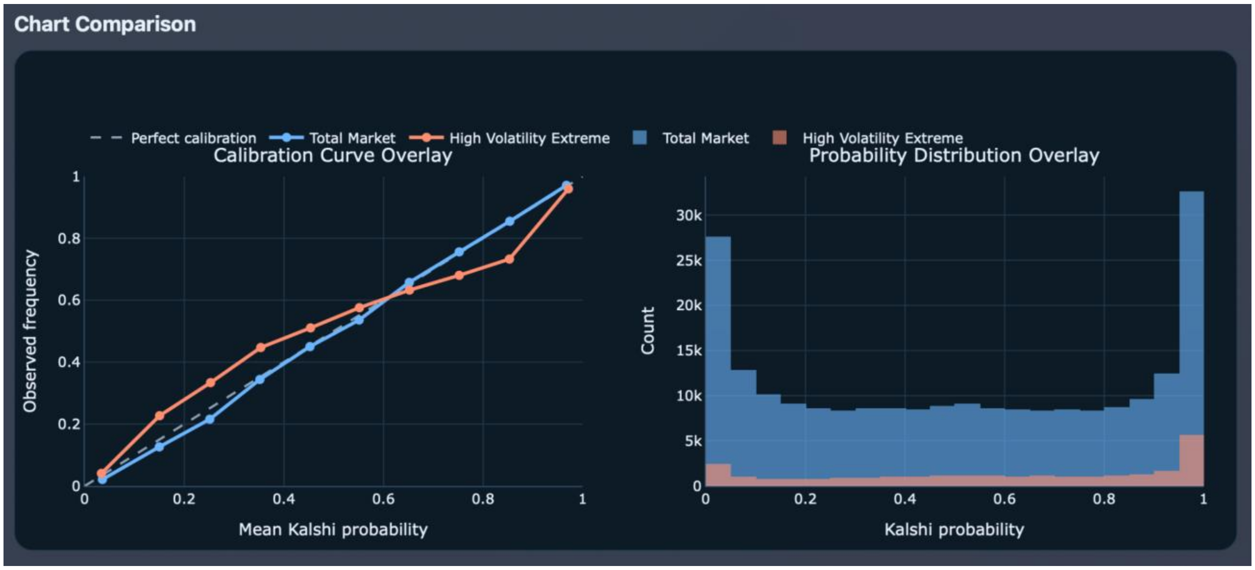
A. **Initial market benchmark:** https://github.com/KingSockss/Binary-Options-Market-Research
B. **Volatility diagnostics and model development:** https://github.com/KingSockss/Volatility-Specific-Probabilistic-Classification-BTC-Model
C. **Classifier:** https://github.com/KingSockss/Volatility-Forecasting-Experimentation
D. **Final Model O orchestration and execution logic:** https://github.com/KingSockss/Model_O_Trading_Strategy_Refinement
E. **Kalshi market candlesticks:** https://docs.kalshi.com/api-reference/market/get-market-candlesticks
F. **Kalshi market order book:** https://docs.kalshi.com/api-reference/market/get-market-orderbook
G. **Kalshi fee schedule:** https://kalshi.com/fee-schedule
H. **Kalshi fee rounding:** https://docs.kalshi.com/getting_started/fee_rounding

## Appendix B: Evaluation Results

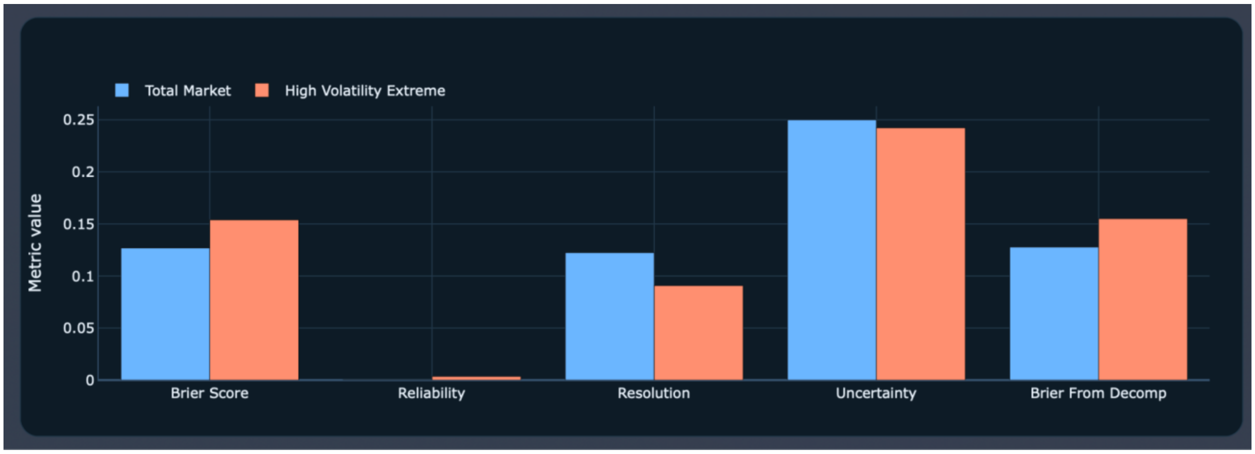
| Plot/Repository Label | Name Used in this Paper | Short Description |
|---|---|---|
| Model A | Independent Model | Classic/normal independent probability model |
| Model K | Kalshi's Market | Kalshi's BTC hourly contracts market interpreted as a probability model |
| Model K RT | Kalshi's Market during HVE | Real-time Kalshi market interpreted as a probability model on only high volatility extreme minutes |
| Model H RT | Volatility Specialized Model | Volatility specialized model trained specifically on high-volatility minutes with added post-hoc layers |

**Model naming note:** The figures retain the original labels used during development and in the repositories. For readability, the main text uses descriptive names. The table above provides mapping between the two naming systems.

A. Kalshi Global vs. HVE Calibration Curve

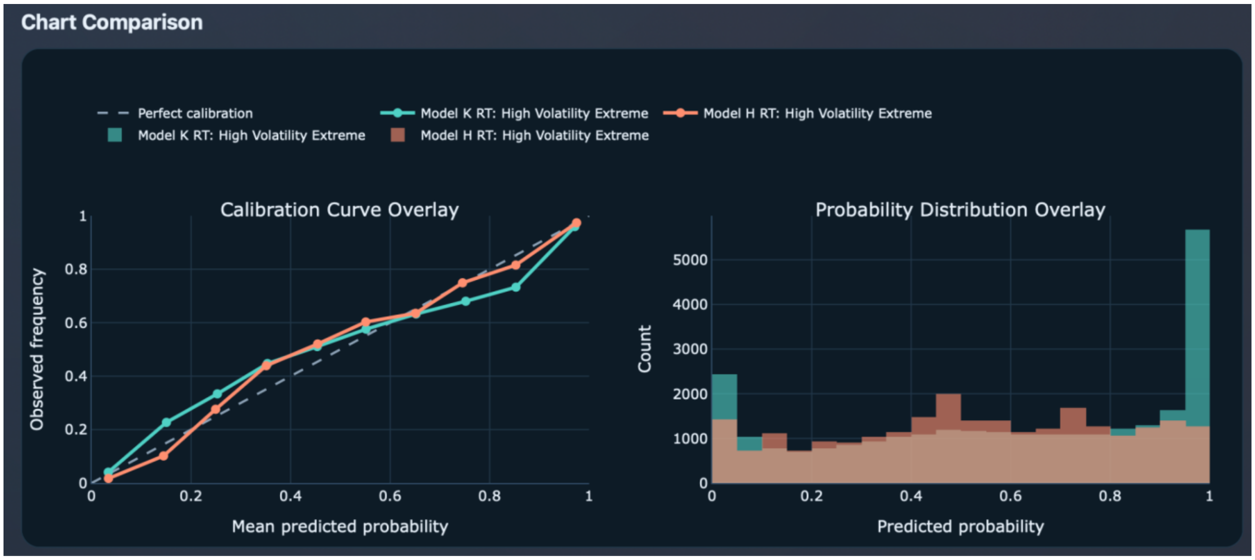


B. Kalshi Global vs. HVE Brier Decomposition

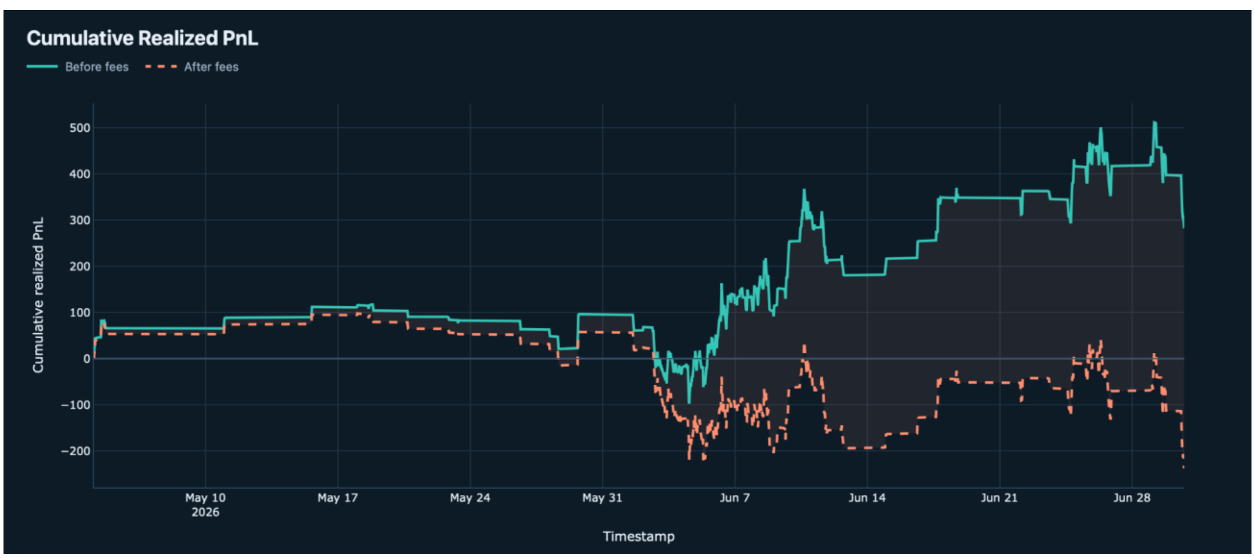



C. Kalshi Global Model vs. Independent Probability Base Model

D. Kalshi Market vs. Volatility-Specific Independent Model on HVE subset

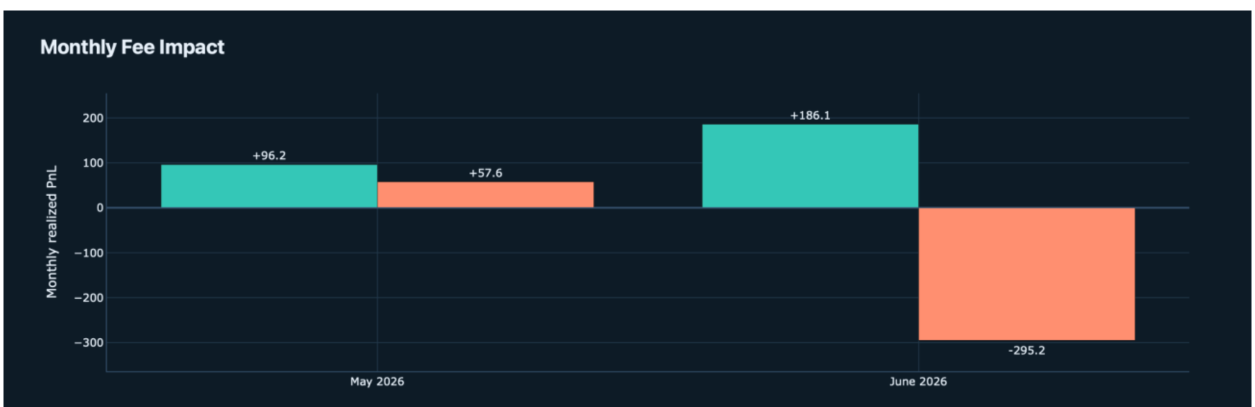



E. Cumulative Realized PnL of Hybrid Model Backtest Gross vs. Net



F. Strategy Drawdown Before/After Fees


G. Strategy Monthly Fee Impact on Backtest Period




| Month | Trades | Before-fee PnL | Fees | After-fee PnL | ROI before fees | ROI after fees |
|---|---|---|---|---|---|---|
| May 2026 | 2,608 | +96.195 | 38.600 | +57.595 | 7.92% | 4.60% |
| June 2026 | 31,246 | +186.070 | 481.250 | -295.180 | 1.39% | -2.13% |